MARKET REQUIREMENTS DOCUMENT (MRD) • DELIVERABLE A

Hospital OP Live Tracker

Multi-Hospital Outpatient Queue Orchestration, AI Waiting-Time Intelligence & Emergency Bed Tracking System

Document Version:	1.0.0 (Approved Baseline)	**Review Status:**	Formal Baseline Approved
Release Date:	October 24, 2026	**Document Owner:**	Principal PM & Healthcare Solutions Architect
Geographic Pilot:	Tiruchirappalli (Trichy), TN, India	**Target Stack:**	React 19 + FastAPI + MongoDB + WebSockets

1. Executive Summary & Product Vision

Outpatient (OP) departments across public and private hospitals face severe operational bottlenecks characterized by chaotic waiting rooms, opaque queue sequencing, predatory token scalping, high patient anxiety, and specialist time wastage. The **Hospital OP Live Tracker** is an enterprise healthcare platform engineered to eliminate waiting room congestion through a single authoritative queue engine combining online token pre-booking with walk-in triage, automated arrival check-ins, autonomous background token expiry, high-contrast digital waiting hall signage, transparent waiting-time intelligence, and real-time emergency bed availability tracking.

2. Problem Statement & Operational Pain Points

- **Chaotic Waiting Halls & Nosocomial Risk:** Patients congregate in dense physical waiting areas for 2 to 4 hours without visibility into when they will be examined, elevating disease transmission risks and patient-staff conflicts.
- **Queue Fraud & Duplicate Bookings:** Unregulated manual paper queues or unindexed booking apps suffer from duplicate token issuance, queue-jumping, and token hoarding by scalpers.
- **Specialist Slot Wastage via Unmanaged No-Shows:** Patients who book tokens but abandon visits without notifying staff waste 15% to 30% of prime consulting slots because traditional systems lack autonomous expiry mechanisms to advance the queue.
- **Opaque Doctor Delays:** When emergency surgeries, rounds, or urgent trauma cases delay a doctor, waiting outpatients receive zero notice, magnifying frustration and crowd agitation.
- **Disconnected Emergency Bed Intelligence:** During regional medical emergencies, patients and 108 ambulance crews waste golden-hour transit time visiting hospitals blindly because real-time ICU and oxygen bed availability is not publicly visible.

3. Target Market & Intended Hospital Segments

The product addresses healthcare delivery institutions across India, with an initial regional validation anchor in **Tiruchirappalli (Trichy) District, Tamil Nadu**, spanning three distinct operational segments:

- **Segment 1: Government General & Teaching Medical Colleges (e.g., MGMGH Trichy, SRM Medical College):** Extreme patient volume (1,500–3,500 OP visits/day), predominantly walk-in registrations, requiring rugged zero-duplicate token generation, dual-tone audio chimes, and high-contrast digital signage.
- **Segment 2: Regional Multi-Specialty Private Hospital Networks (e.g., Kauvery Cantonment & Heartcity, Apollo Speciality, Frontline Hospital):** Driven by patient experience, slot punctuality, mobile OTP authentication, and transparent wait-time forecasting to reduce waiting lounge crowding.
- **Segment 3: Charitable & Mission Hospitals (e.g., CSI Mission Hospital):** Low-friction technology adoption with dual-engine offline resilience and shared emergency bed reporting.

4. Stakeholder Groups, Goals & Adoption Barriers

Stakeholder	Primary Goal	Key Functional Need	Adoption Barrier & Mitigation
Patient & Caregiver	Minimize physical wait time; know exact status	Passwordless OTP booking, live queue position, travel alerts	Digital literacy: simplified UI, walk-in parity, audio chimes
OP Registration Staff	Process walk-in crowds without duplicate errors	Fast walk-in desk booking, automated check-in confirmation	Workload friction: single-click 5-second walk-in issuance
Nurses & OPD Staff	Maintain orderly queue flow outside consulting rooms	Queue status screen, patient attendance tracking, hold queue	Cluttered software: clean tablet-optimized queue manager
Consulting Doctor	Focus on patient care without administrative friction	Single-click console (Call, Start, Complete), delay broadcast	Time loss: 1-tap delay broadcast (+15m/+30m) with live push
Hospital Admin	Maximize room utilization; reduce waiting room chaos	Live throughput analytics, bed census update, policy config	System downtime: dual-engine offline memory store resilience
Platform Admin	Oversee regional health network and tenant integrity	Multi-tenant onboarding, cross-hospital isolation, health API	Security concerns: role-based access control, tenant scoping

5. Value Propositions by Stakeholder

- **For Patients:** Zero wasted hospital hours. Book from home, receive travel departure recommendations based on traffic, view live queue movement, and avoid arriving hours before the doctor is ready.
- **For Doctors:** Continuous, orderly patient throughput. No unannounced walk-ins barging into consulting rooms. Single-click delay broadcasting automatically adjusts all patient reporting deadlines when emergency surgeries arise.
- **For Staff:** Elimination of queue arguments. Server-authoritative token numbers, automated auto-expiry for no-shows, and digital waiting hall signage ensure fair, transparent patient calling.
- **For Administrators:** Actionable throughput velocity metrics, recovered appointment slots, optimized staff allocation, and regional emergency bed leadership.

6. Market-Level Product Requirements & Priorities

Req ID	Market / User Need	Proposed Product Capability	Priority	Validation Status
MR-001	Frictionless patient identity without passwords	Mobile OTP authentication with auto-provisioning	Must Have	**Verified in Repo**
MR-002	Discover hospital services and doctor schedules	Multi-hospital directory, doctor OPD schedules	Must Have	**Verified in Repo**
MR-003	Pre-book outpatient tokens with arrival deadlines	Atomic online booking with reporting deadline	Must Have	**Verified in Repo**
MR-004	Equal access for walk-in and online outpatients	Unified queue engine with zero duplicate token guarantee	Must Have	**Verified in Repo**
MR-005	Recover unused consultation slots from no-shows	1-tap arrival check-in and automated background expiry	Must Have	**Verified in Repo**
MR-006	Consulting room flow and delay communication	Doctor console (Call, Visit, Delay broadcast push)	Must Have	**Verified in Repo**
MR-007	Manage hospital departments, staff, and policies	Hospital administration and OP schedule configuration	Should Have	Partially Implemented
MR-008	Platform-level multi-tenant health network	Platform admin onboarding with tenant data isolation	Must Have	Planned (Spec Defined)
MR-009	Real-time critical bed visibility for ambulances	Emergency bed tracker (ICU, Trauma, O2) with status	Must Have	**Verified in Repo**
MR-010	Transparent waiting time expectation management	Transparent heuristic wait forecast + ML regression hooks	Must Have	**Verified in Repo**
MR-011	Instant status alerts across multiple channels	Real-time WebSockets hub + multi-event SMS/in-app alerts	Must Have	**Verified in Repo**
MR-012	Accessible waiting hall calling for elderly/ill	Waiting Hall TV mode signage with dual-tone audio chime	Must Have	**Verified in Repo**
MR-013	Travel guidance to prevent missing OP deadlines	Trichy locality travel time calculation with rush hour buffer	Should Have	Verified Heuristic
MR-014	Operational visibility into clinic throughput	Hourly outpatient arrival/completion velocity charts	Should Have	**Verified in Repo**

7. Scope, Exclusions & Strategic Roadmap

In-Scope: Multi-hospital directory; patient mobile OTP authentication; atomic online OP token reservation; walk-in OP desk registration; unified FIFO queue ordering; 1-tap arrival check-in; autonomous background expiry worker with crash recovery; doctor consultation console; doctor OPD delay broadcast push; high-contrast waiting hall TV signage with Web Audio chimes; baseline heuristic wait-time prediction; emergency bed status dashboard; and regional travel time calculation.

- **No Clinical Diagnosis / Medical Triage:** The system does not provide medical diagnostic recommendations, prescribe drugs, or substitute for emergency 108 triage.
- **No Inpatient / Billing Integration:** Does not process hospital inpatient ward admissions, complex insurance claim reimbursements, or dispensary inventory.
- **No Biometric Hardware Mandate:** Aadhaar e-KYC biometric fingerprint/iris scanners are excluded; system operates with phone OTP and masked identity inputs in compliance with Indian privacy laws.

8. Success Indicators & Target Metrics

Key Performance Indicator (KPI)	Target Operational Benchmark	Measurement & Verification Method
Waiting Room Peak Crowd Density	> 40% reduction in physical waiting crowd	Pre- and post-pilot waiting lounge headcounts
No-Show Slot Recovery Rate	> 70% of abandoned slots recycled to queue	Audit log tracking of EXPIRED tokens advancing next patient
Wait Time Forecast Accuracy	Within ±15% of actual consultation call time	Comparison of predicted wait minutes vs actual call timestamps
Zero Duplicate Token Incidents	100% mathematical integrity (0 duplicates)	Compound database unique constraint verification & stress tests
Emergency Bed Update Latency	< 30 minutes from physical bed status change	Audit timestamps of bed update API calls against hospital census

9. High-Level Product Roadmap

- **Phase 1: Foundation & Core Queue Prototype (Current Status):** Scaffolded React 19 frontend and FastAPI backend; verified in-memory store; verified atomic booking, check-in, background auto-expiry worker, TV display, and 11 automated pytest test cases.
- **Phase 2: Pilot Hospital Department Hardening (Weeks 1–4):** Deploy in single OPD room (Cardiology at Apollo/Kauvery Trichy); validate doctor console UX; connect live MongoDB Atlas cluster; integrate transactional SMS gateway with DLT registered templates.
- **Phase 3: Multi-Hospital & Emergency Bed Expansion (Weeks 5–8):** Onboard all 6 Trichy partner hospitals; operationalize emergency bed tracking with ward nurse mobile interface; deploy public waiting hall screens.
- **Phase 4: AI Optimization & ABDM Integration (Weeks 9–16):** Train scikit-learn regression models on collected consultation timestamps; integrate Ayushman Bharat Health Account (ABHA) patient identity; support WhatsApp Business conversational booking.

10. Stakeholder Validation & Operational Pilot Plan

- **Stage 1: Synthetic Simulation:** Validate concurrency with 30+ simultaneous booking threads and test background worker catch-up after simulated server crashes (Verified in test suite).
- **Stage 2: Single-Clinic Shadow Pilot:** Deploy the Doctor Console and Waiting Hall TV screen in a single outpatient clinic for 5 consecutive days without modifying hospital administrative records to assess doctor adherence and chime clarity.
- **Stage 3: Full OP Department Pilot:** Activate live patient booking with SMS dispatch and walk-in integration across General Medicine and Cardiology departments with weekly stakeholder retrospectives.